

Command line seminar-2

Bioinformatics course
Term 2, 2024
Skoltech

Marina Pak

Writing and running scripts



Running long processes: scripts and screen

? How to make a script?

? How to use screen?

Running long processes: scripts and screen



How to make a script?



How to use screen?



Bash scripting

1. The first line in the script should be `#!/bin/bash`. It tells the interpreter that the script is written in Bash.
2. Write your commands from the new lines. Or use chaining with `;`.
3. Save the script with the extension `.sh`.
4. Make the script executable:
`chmod +x script.sh`.
5. Run the script: `./script.sh`.

Running long processes: scripts and screen



How to make a script?



Bash scripting

1. The first line in the script should be `#!/bin/bash`. It tells the interpreter that the script is written in Bash.
2. Write your commands from the new lines. Or use chaining with `;`.
3. Save the script with the extension `.sh`.
4. Make the script executable:
`chmod +x script.sh`.
5. Run the script: `./script.sh`.



How to use screen?



screen

Create a screen: `screen -S [screen_name]`

Exit screen: `Ctrl + a, d` or `screen -d`

Resume screen: `screen -r [screen_name]`

List of screens: `screen -ls`

Kill screen: `screen -k`

Running long processes: scripts and screen

- ? Write a script imitating a long-running process using a command `sleep` and run it in the `screen`.

Running long processes: scripts and screen

- ? Write a script imitating a long-running process using a command `sleep` and run it in the `screen`.

```
#!/bin/bash  
echo 'Running some heavy computations...'  
sleep 300s
```

Running long processes: scripts and screen

- ? Write a script imitating a long-running process using a command `sleep` and run it in the `screen`.

```
#!/bin/bash  
echo 'Running some heavy computations...'  
sleep 300s
```

- ? Modify the script to accept an argument with the time for a sleep.

Running long processes: scripts and screen

- ? Write a script imitating a long-running process using a command **sleep** and run it in the **screen**.

```
#!/bin/bash  
echo 'Running some heavy computations...'  
sleep 300s
```

- ? Modify the script to accept an argument with the time for a sleep.

```
#!/bin/bash  
echo 'Running some heavy computations...'  
echo 'Estimated time: $1'  
sleep $1
```

Running long processes: scripts and screen

- ? Write a script imitating a long-running process using a command `sleep` and run it in the `screen`.

```
#!/bin/bash  
echo 'Running some heavy computations...'  
sleep 300s
```

- ? Modify the script to accept an argument with the time for a sleep.

```
#!/bin/bash  
echo 'Running some heavy computations...'  
echo 'Estimated time: $1'  
sleep $1
```

- ? Save script output to log.

Running long processes: scripts and screen

- ? Write a script imitating a long-running process using a command `sleep` and run it in the `screen`.

```
#!/bin/bash  
echo 'Running some heavy computations...'  
sleep 300s
```

- ? Modify the script to accept an argument with the time for a sleep.

```
#!/bin/bash  
echo 'Running some heavy computations...'  
echo 'Estimated time: $1'  
sleep $1
```

- ? Save script output to log.
`./script.sh > log.txt`



About logging

- Log is a good practice. Use it in your scripts.
- Well-written programs save detailed human-readable log.
- Sometimes you have to work with bad-written programs.
- Redirecting script output to a file does not save errors and warnings.
- To log all messages from commands, add four lines into your script (see Summary section in this post:
<https://blog.tratif.com/2023/01/09/bash-tips-1-logging-in-shell-scripts/>).
- For advanced scripting better use Python.

Break time!



Writing universal scripts



RMSD:
5.509Å
TM-score:
0.50220

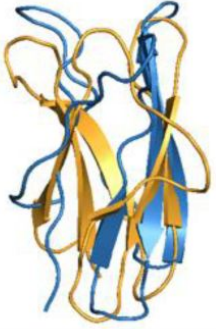


RMSD:
0.700Å
TM-score:
0.92300

From <https://doi.org/10.3390/ijms24076262>

Writing universal scripts

For the next tasks you will need a program for structural alignment TMalign.
Download it by the link https://zhanggroup.org/TM-align/TMalign_cpp.gz.



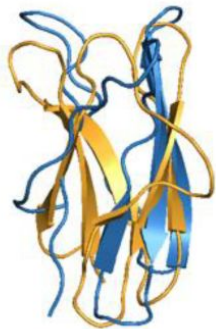
RMSD:
5.509Å
TM-score:
0.50220



RMSD:
0.700Å
TM-score:
0.92300

From <https://doi.org/10.3390/ijms24076262>

Writing universal scripts



RMSD:
5.509Å
TM-score:
0.50220



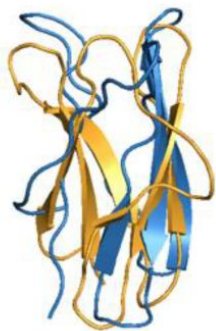
RMSD:
0.700Å
TM-score:
0.92300

For the next tasks you will need a program for structural alignment TMalign. Download it by the link https://zhanggroup.org/TM-align/TMalign_cpp.gz.

?

Write a script that takes two protein structure files and saves the TM-score metrics of TMalign alignment to a file named by the names of the two input structure files.

Writing universal scripts



RMSD:
5.509Å
TM-score:
0.50220



RMSD:
0.700Å
TM-score:
0.92300

From <https://doi.org/10.3390/ijms24076262>

For the next tasks you will need a program for structural alignment TMalign. Download it by the link https://zhanggroup.org/TM-align/TMalign_cpp.gz.



Write a script that takes two protein structure files and saves the TM-score metrics of TMalign alignment to a file named by the names of the two input structure files.



Hints

- Bioinformatics tools are usually ready-to-use from the archive. If you get an error when running a program, make it executable with `chmod +x [program_file]`.
- Figure out how to submit files to TMalign – read the manual.
- Run TMalign with two structure files, see how the output looks like and what do you need from it.
- Write an “applied” command that takes two files and does the required actions.
- Wrap the command into the script. Make the script universal by replacing concrete filenames with variables.

Writing universal scripts: for-loop



Modify the script from the previous task to align all structures from the directory `/home/Shared/models/` against a reference structure `/home/Shared/pdbs/mor.pdb`. A reference structure is supplied to TMalign as the first structure.

Your feedback!

Feel the feedback form
(link will follow).

